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Git CI/CD

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What is CI/CD?

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CI/CD stands for **Continuous Integration** and **Continuous Deployment/Delivery**.

It means your code is automatically tested and deployed every time you push.

This helps you catch bugs early and deliver features faster, with less manual work.

Why Use CI/CD?

CI/CD automates the process of testing and deploying your code. This means:

- Find bugs before they reach users
- Deploy changes faster and more safely
- Reduce manual steps and mistakes
- Get quick feedback on every push

How Does CI/CD Work with Git?

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- If something fails, you get notified right away

Example Workflow

```
[Developer] --push--> [Git Repository] --triggers--> [CI/CD Pipeline: Test, Build, Deploy]
```

Popular CI/CD Services

- **GitHub Actions:** Built into GitHub, uses YAML files in `.github/workflows/`
- **GitLab CI/CD:** Built into GitLab, uses `.gitlab-ci.yml`
- **CircleCI:** Works with GitHub/GitLab, easy setup for many languages
- **Travis CI:** Popular for open-source, uses `.travis.yml`
- **Azure Pipelines:** Works with Azure DevOps and GitHub, supports many platforms

Key CI/CD Concepts

Here are some important terms:

- **Workflow:** A series of jobs that run together
- **Job:** A group of steps that run together
- **Step:** A single task, like checking out code or running tests
- **Runner:** The computer/server that runs your jobs
- **Trigger:** Decides when your workflow runs
- **Environment Variables:** Settings for your workflow
- **Secrets:** Passwords or API keys

Jobs

A **job** is a group of steps that run together. Each job runs on a runner (a server).

Example: A Job in GitHub Actions

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```
steps:
  # steps go here
```

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Steps

Each **step** is a single task, like checking out code or running tests.

Example: Steps

```
steps:
  - uses: actions/checkout@v3
  - name: Run tests
    run: npm test
```

Runners

A **runner** is the computer/server that runs your jobs.

You can use the service's runners or set up your own for more control.

Example: Specify a Runner

```
runs-on: ubuntu-latest
```

Triggers

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Example: Trigger on Push or Pull Request

```
on:
  push:
  pull_request:
```

Environment Variables & Secrets

Use **environment variables** for settings, and **secrets** for passwords or API keys.

Never hardcode secrets in your code!

Example: Use a Secret

```
env:
  NODE_ENV: production
  API_KEY: ${{ secrets.API_KEY }}
```

Build Logs

CI/CD tools show logs for every job and step. Check logs to see what happened or to debug failures.

In GitHub Actions, click on a workflow run and see logs for each job/step.

Skipping CI

You can skip CI/CD for a commit by adding `[skip ci]` to your commit message.

This is useful for documentation or minor changes.

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Badges

Add a badge to your README to show CI/CD status.

This lets others see if your latest build passed.

Example: GitHub Actions Badge

```
![CI](https://github.com/username/repo/actions/workflows/ci.yml/badge.svg)
```

Example: GitHub Actions Workflow File (Explained)

```
# .github/workflows/ci.yml
# This file tells GitHub Actions how to run CI for your project

name: CI                  # The name of the workflow (shows up in GitHub)
on: [push]                # Trigger: run this workflow on every push
jobs:
  build:                  # Job name (can be anything)
    runs-on: ubuntu-latest   # Runner: use the latest Ubuntu server
    steps:
      - uses: actions/checkout@v3  # Step: check out your code from the repo
      - name: Run tests            # Step: give this step a name
        run: npm test              # Step: run your project's tests
```

- **name:** Sets the workflow's display name in GitHub.
- **on:** Decides when the workflow runs (here: every push).
- **jobs:** Groups together steps that run on a runner.
- **build:** The name of this job (can be anything).

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- **run:** Runs a shell command (here: `npm test` to run tests).

Troubleshooting & Best Practices

- If a build fails, check the logs for error messages.
- Make sure your secrets and environment variables are set correctly.
- You can rerun failed jobs from the CI/CD dashboard.
- Check the documentation for your CI/CD service for more help.
- Start small: automate tests first, then add deployment when ready.
- Keep secrets out of your code and never commit API keys.
- Use badges to show your build status in the README.

Note: CI/CD helps catch bugs early and speeds up delivery. Even small projects can benefit from automation!

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